

Project Part 1 exam: Memory Management: Victim Replacement Algorithms

Instructions

- **MAKE SURE TO USE THE EXAM VERSION OF THE SIMULATOR**
SimVictimExam.py
- EDIT THIS document to insert your answers
- Time: 40mins - Submit your completed document on the submission link provided. DO NOT submit a PDF.
- Where necessary copy and paste images of the data in question throughout your document to illustrate your answers.
- If necessary, re-run the program but make sure to show the new data in your answer.

Paste the ID and date and timestamp from the simulator here:

```
Thank you R00225669 of SD2A 23/10/2024, 09:45:31
NOTE
```

```
Please include a snip of this ID number,
group name and timestamp in your submitted document.

And, when they appear, also please include later snips
of the ID number, group name and timestamp in your submitted document.
```

Answer both questions

Run each algorithm listed below in turn. For each one, copy and paste **ALL** of the output and give the explanations indicated. **In all cases use examples from the data generated to illustrate you answers. Explanations that do not refer to the data are not acceptable.**

```
RAM is now full and stores 12 pages as follows:
Frame Page Arrival Time
0      794      0
1      741      1
2      759      2
3      785      5
4      827      6
5      745      7
6      770      8
7      795     10
8      740     11
9      758     15
10     813     16
11     829     26

A reference to page 999 generates a page fault at time 34 so must be installed in RAM.
RAM is full. Therefore a victim page must be chosen from the frames in RAM.

1: Run First In, First Out (FIFO) Frame Victim Replacement Algorithm
2: Run Not Recently Used (NRU) Frame Victim Replacement Algorithm
3: Run Least Recently Used (LRU) Frame Victim Replacement Algorithm
4: Run Least Frequently Used (LFU) Frame Victim Replacement Algorithm
5: Run Least Frequently Used with Decaying Counter (LFUDC) Frame Victim Replacement Algorithm
6: Run Least Recently Used with Ageing Counter (LRUA) Frame Victim Replacement Algorithm
7: Run Clock (aka Second Chance) Frame Victim Replacement Algorithm
8: Generate New Random Data
0: Quit
Which action (0 to 8)?3
```

1. LRU (30 marks)

Fully explain why the victim chosen was picked **with reference to the specific data used**.

```
***** Least Recently Used *****

Reference recency summary, most recent page referenced is shown last: [795, 740, 745, 813, 794, 829, 759, 827, 758, 785, 770, 741, 999]

List of page numbers referenced before new page entered is as follows (most recent reference last):
[794, 741, 759, 794, 794, 785, 827, 745, 770, 827, 795, 740, 741, 740, 745, 758, 813, 794, 770, 827, 770, 785, 785, 827, 759, 827, 829, 827, 759, 827, 758,
785, 770, 741, 999]

There are 12 pages in RAM (assume RAM is now full)

Frames in SD2A R00225669 memory are shown in frame number order.

Frames in SD2A R00225669 23/10/2024, 09:48:54 memory before victim replacement were:
Frame Page arrival time reference bit modify bit reference counter
0 794 0 0 0 4 = 00000100
1 741 1 1 0 3 = 00000011
2 759 2 0 0 3 = 00000011
3 785 5 1 0 4 = 00000100
4 827 6 1 0 7 = 00000111
5 745 7 0 1 2 = 00000010
6 770 8 1 1 4 = 00000100
7 795 10 0 0 1 = 00000001
8 740 11 0 1 2 = 00000010
9 758 15 1 0 2 = 00000010
10 813 16 0 1 1 = 00000001
11 829 26 0 0 1 = 00000001

Press <ENTER> to see which page was chosen as victim...3|

SD2A R00225669 Victim chosen was: page 795 at 23/10/2024, 09:48:54

Page 999 was inserted instead, so:

Frames in memory are now:
Frame Page arrival time reference bit modify bit reference counter
0 794 0 0 0 4 = 00000100
1 741 1 1 0 3 = 00000011
2 759 2 0 0 3 = 00000011
3 785 5 1 0 4 = 00000100
4 827 6 1 0 7 = 00000111
5 745 7 0 1 2 = 00000010
6 770 8 1 1 4 = 00000100
7 999 34 1 0 1 = 00000001
8 740 11 0 1 2 = 00000010
9 758 15 1 0 2 = 00000010
10 813 16 0 1 1 = 00000001
11 829 26 0 0 1 = 00000001

Press <ENTER> to continue
```

Least Recently Used, when picking a victim, the page with the longest amount of time that hasn't been used is the one that will be replaced. In this case the victim chosen was page 795, Page 795 because it has a reference bit of 0, modify bit of 0 and arrival time of 10 meaning that it is the one that has not been used in the longest amount of time. Because of this page 795 is swapped out for page 999.

2. LRU Ageing (70 marks)

- i. **Fully explain** why the victim chosen was picked **with reference to the specific data used**

```
Which action (0 to 8)?SD2A6
Choose an option between 0 and 8
Which action (0 to 8)?6

***** Least Recently Used with Ageing*****

First we will show the ageing history
Press <ENTER> to continue to the history
Reference counter ageing history
In 5 references up to time 5 the pages referenced pages were: [794, 741, 759, 794, 794]
And all references so far are: [794, 741, 759, 794, 794]

After 1 ageing (s) the current reference history is:
Frame  Page  Reference count Reference History Byte  Reference bit
0       794      3             10000000 =128          1
1       741      1             10000000 =128          1
2       759      1             10000000 =128          1
3       785      0             00000000 =0           0
4       827      0             00000000 =0           0
5       745      0             00000000 =0           0
6       770      0             00000000 =0           0
7       795      0             00000000 =0           0
8       740      0             00000000 =0           0
9       758      0             00000000 =0           0
10      813      0             00000000 =0           0
11      829      0             00000000 =0           0
Press <ENTER> to continue to next part of the history
In 5 references up to time 10 the pages referenced pages were: [785, 827, 745, 770, 827]
And all references so far are: [794, 741, 759, 794, 794] [785, 827, 745, 770, 827]

After 2 ageing (s) the current reference history is:
Frame  Page  Reference count Reference History Byte  Reference bit
0       794      3             01000000 =64          0
1       741      1             01000000 =64          0
2       759      1             01000000 =64          0
3       785      1             10000000 =128         1
4       827      2             10000000 =128         1
5       745      1             10000000 =128         1
6       770      1             10000000 =128         1
After 2 ageing (s) the current reference history is:
Frame  Page  Reference count Reference History Byte  Reference bit
0       794      3             01000000 =64          0
1       741      1             01000000 =64          0
2       759      1             01000000 =64          0
3       785      1             10000000 =128         1
4       827      2             10000000 =128         1
5       745      1             10000000 =128         1
6       770      1             10000000 =128         1
7       795      0             00000000 =0           0
8       740      0             00000000 =0           0
9       758      0             00000000 =0           0
10      813      0             00000000 =0           0
11      829      0             00000000 =0           0
Press <ENTER> to continue to next part of the history
In 5 references up to time 15 the pages referenced pages were: [795, 740, 741, 740, 745]
And all references so far are: [794, 741, 759, 794, 794] [785, 827, 745, 770, 827] [795, 740, 741, 740, 745]

After 3 ageing (s) the current reference history is:
Frame  Page  Reference count Reference History Byte  Reference bit
0       794      3             00100000 =32          0
1       741      2             10100000 =160         1
2       759      1             00100000 =32          0
3       785      1             01000000 =64          0
4       827      2             01000000 =64          0
5       745      2             11000000 =192         1
6       770      1             01000000 =64          0
7       795      1             10000000 =128         1
8       740      2             10000000 =128         1
9       758      0             00000000 =0           0
10      813      0             00000000 =0           0
11      829      0             00000000 =0           0
Press <ENTER> to continue to next part of the history
In 5 references up to time 20 the pages referenced pages were: [758, 813, 794, 770, 827]
And all references so far are: [794, 741, 759, 794, 794] [785, 827, 745, 770, 827] [795, 740, 741, 740, 745] [758, 813, 794, 770, 827]
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After 4 ageing (s) the current reference history is:
Frame Page Reference count Reference History Byte Reference bit
0 794 4 10010000 =144 1
1 741 2 01010000 =80 0
2 759 1 00010000 =16 0
3 785 1 00100000 =32 0
4 827 3 10100000 =160 1
5 745 2 01100000 =96 0
6 770 2 10100000 =160 1
7 795 1 01000000 =64 0
8 740 2 01000000 =64 0
9 758 1 10000000 =128 1
10 813 1 10000000 =128 1
11 829 0 00000000 =0 0
Press <ENTER> to continue to next part of the history
In 5 references up to time 25 the pages referenced pages were: [770, 785, 785, 827, 759]
And all references so far are: [794, 741, 759, 794, 794] [785, 827, 745, 770, 827] [795, 740, 741, 740, 745] [758, 813, 794, 770, 827] [770, 785, 785, 827, 759]

After 5 ageing (s) the current reference history is:
Frame Page Reference count Reference History Byte Reference bit
0 794 4 01001000 =72 0
1 741 2 00101000 =40 0
2 759 2 10001000 =136 1
3 785 3 10010000 =144 1
4 827 4 11010000 =208 1
5 745 2 00110000 =48 0
6 770 3 11010000 =208 1
7 795 1 01000000 =32 0
8 740 2 00100000 =32 0
9 758 1 01000000 =64 0
10 813 1 01000000 =64 0
11 829 0 00000000 =0 0
Press <ENTER> to continue to next part of the history
In 5 references up to time 30 the pages referenced pages were: [827, 829, 827, 759, 827]
And all references so far are: [794, 741, 759, 794, 794] [785, 827, 745, 770, 827] [795, 740, 741, 740, 745] [758, 813, 794, 770, 827] [770, 785, 785, 827, 759] [827, 829, 827, 759, 827]

After 6 ageing (s) the current reference history is:
Frame Page Reference count Reference History Byte Reference bit
0 794 4 00100100 =36 0
1 741 2 00010100 =20 0
2 759 3 11000100 =196 1
3 785 3 01001000 =72 0
4 827 7 11101000 =232 1
5 745 2 00011000 =24 0
6 770 3 01101000 =104 0
7 795 1 00010000 =16 0
8 740 2 00010000 =16 0
9 758 1 00100000 =32 0
10 813 1 00100000 =32 0
11 829 1 10000000 =128 1
Press <ENTER> to continue to next part of the history
In 5 references up to time 34 the pages referenced pages were: [758, 785, 770, 741]
And all references so far are: [794, 741, 759, 794, 794] [785, 827, 745, 770, 827] [795, 740, 741, 740, 745] [758, 813, 794, 770, 827] [770, 785, 785, 827, 759] [827, 829, 827, 759, 827] [758, 785, 770, 741]

After 7 ageing (s) the current reference history is:
Frame Page Reference count Reference History Byte Reference bit
0 794 4 00010010 =18 0
1 741 3 10001010 =138 1
2 759 3 01100010 =98 0
3 785 4 10100100 =164 1
4 827 7 01110100 =116 0
5 745 2 00001100 =12 0
6 770 4 10110100 =180 1
7 795 1 00001000 =8 0
8 740 2 00001000 =8 0
9 758 2 10010000 =144 1
10 813 1 00010000 =16 0
11 829 1 01000000 =64 0
Press <ENTER> to continue to next part of the history

Reference recency summary, most recent page referenced is shown last: [795, 740, 745, 813, 794, 829, 759, 827, 758, 785, 770, 741, 999]

List of page numbers referenced before new page entered is as follows (most recent reference last):
[794, 741, 759, 794, 794, 785, 827, 745, 770, 827, 795, 740, 741, 740, 745, 758, 813, 794, 770, 827, 770, 785, 785, 827, 759, 827, 829, 827, 759, 827, 758, 785, 770, 741, 999]

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There are 12 pages in RAM (assume RAM is now full)
Frames in SD2A R00225669 memory are shown in frame number order.

Frames in SD2A R00225669 23/10/2024, 10:04:16 memory before victim replacement were:
Frame  Page  arrival time  reference bit  modify bit  reference counter  reference history
0      794    0            0            0            4      = 00000100    00010010 = 18
1      741    1            1            0            3      = 00000011    10001010 = 138
2      759    2            0            0            3      = 00000011    01100010 = 98
3      785    5            1            0            4      = 00000100    10100100 = 164
4      827    6            0            0            7      = 00000111    01110100 = 116
5      745    7            0            1            2      = 00000010    00001100 = 12
6      770    8            1            1            4      = 00000100    10110100 = 180
7      795   10            0            0            1      = 00000001    00001000 = 8
8      740   11            0            1            2      = 00000010    00001000 = 8
9      758   15            1            0            2      = 00000010    10010000 = 144
10     813   16            0            1            1      = 00000001    00010000 = 16
11     829   26            0            0            1      = 00000001    01000000 = 64

Press <ENTER> to see which page was chosen as victim...

SD2A R00225669 Victim chosen was: page 795 at 23/10/2024, 10:04:16

Page 999 was inserted instead, so:

Frames in memory are now:
Frame  Page  arrival time  reference bit  modify bit  reference counter  reference history
0      794    0            0            0            4      = 00000100    00010010 = 18
1      741    1            1            0            3      = 00000011    10001010 = 138
2      759    2            0            0            3      = 00000011    01100010 = 98
3      785    5            1            0            4      = 00000100    10100100 = 164
4      827    6            0            0            7      = 00000111    01110100 = 116
5      745    7            0            1            2      = 00000010    00001100 = 12
6      770    8            1            1            4      = 00000100    10110100 = 180
7      999   34            1            0            1      = 00000001    10000000 = 128
8      740   11            0            1            2      = 00000010    00001000 = 8
9      758   15            1            0            2      = 00000010    10010000 = 144
10     813   16            0            1            1      = 00000001    00010000 = 16
11     829   26            0            0            1      = 00000001    01000000 = 64

Press <ENTER> to continue

```

Page 795 was picked as the victim because Least Recently Used with Ageing, picks who gets swapped out by what page was used the longest time ago, but it also takes into account how long ago the page was reference, with a reference counter. In this case page 795 had the lowest reference of 8 = 00001000. (Reference counter 1 = 00000001) So it is swapped out for page 999.

ii. **Fully explain the meaning** of the reference history byte values as used in LRU Ageing **using some examples from your data**

The reference history byte shows if a page has been referenced. Keeping track of how long ago a page was referenced. All pages start with a value of 0 (00000000) once referenced its left most bit will turn to 1, (10000000) and will then be halved every time it's not referenced. If the page is referenced it will receive another 1 at its left most bit for example after 3 ageing page 741 gets referenced so its reference history byte is 10100000 (=160) the values keep decrementing by 2 (moved to the right) but every time a page is reference the least most bit changes to 1.

iii. **Fully explain** how the ageing of the reference history byte works using an example from **your** output after 3 ageings

Page 795 has a reference history Byte of 00000000 after 1 ageing,

After 2 ageing it stays the same as it has not been referenced.

After 3 ageing it is referenced so its left most byte goes to 1, 10000000.

After 4 Ageing it is not referenced so its reference byte is moved one

place to the right, decrementing it by 2, so its reference history byte is now 01000000.

After 5 Ageing it is not referenced again to its reference byte is moved one place to the right, decrementing it by 2, so its reference history byte is now 00100000.

After 6 Ageing it is not referenced again to its reference byte is moved one place to the right, decrementing it by 2, so its reference history byte is now 00010000.

After 7 Ageing it is not referenced again to its reference byte is moved one place to the right, decrementing it by 2, so its reference history byte is now 00001000.

At 3 Ageing page 741 gets referenced so its reference history byte is 10100000 (=160) the values keep decrementing by 2 (moved to the right) but every time a page is reference the least most bit changes to 1.

If page 795 was to be referenced (its not but if) After ageing it is referenced so its left most byte goes to 1 so its reference history byte would become 10000100